Programme Name – B. TECH(CSE)

Course Name – Probability & Statistics

Course Code – BBS00014

(Semester – III)

Module III

Content

SL.No	Topic	Pg.No
1	Sample questions and solutions	3-25

Sample questions:

1.	Identify the mode of a binomial distribution with p=1/2 and n=7		3
	a. 3,4	b. 4,5	
	c. 4	d. 5	
2.	Select by which the standard normal distribution is represented		3
	a. N(0,0)	b. N(0,1)	
	c. N(1,1)	d.N(1,0)	
3.	Identify the variance of Geometric Distribution is		3
	a. q/p^2	b. q^2/p	
	c. $p(1-p)/q^2$	d. $q(1-q)/p^2$	
4.	The normal distribution is a limiting form of Binomial distribution, Identify the correct option:		3
	a. n -> ∞ and p ->0	b. n ->0 and p ->∞	
	c. n -> ∞ and p ->n	d. n -> ∞ and neither p nor q is small	
5.	The mean of Bin(n,p) distribution is,Select the correct option.		3
	a. n	b. npq	
	c.n(n-1)p	d.np	
6.	Identify which of the following is not a property of a binomial experiment?		3
	a. The experiment consists of a sequence of n identical trials	b. Each outcome can be referred to as a success or a failure	
	c.the probabilities of the two outcomes can change from one trial to the next	d. the trials are independent	
7.	The rate of accidents in a road is 6 per 14 days. what is the probability that there occur 2 accident in a week. Identify the correct option.		3
	a. 0.22	b. 0.25	
	c. 0.23	d. None of these	
8.	The variance of Uniform distribution is , Select the correct option		3
	a. $\frac{b-a}{12}$	b. $\frac{(b-a)^2}{12}$	

	c. $\frac{a+b}{2}$	d. None of these.	
9.	If E(X)=3, E[X(X-1)] =22, then estimate the value of Var(7-2x)		3
	a. 16	b. 64	
	c. 32	d. None of these.	
10.	Let X follows Normal distribution with mean 45 and standard deviation 5 and ϕ(1)=0.84,then estimate P(X≤40) is		3
	a. 0.16	b. 0.84	
	c. 0.68	d. None of these.	
11.	Estimate the mode(s) of a random variable X following a binomial distribution with parameters m=10,p=0.7		3
	a. 6	b. 6,7	
	c. 7	d. 7,8	
12.	Identify the shape of a binomial distribution with parameters 5 and 1/3 is,		3
	a. leptokurtic	b. platykurtic	
	c. mesokurtic	d. none of these	
13.	Bernoullian trials are. Select the correct option.		3
	a. Dependent	b. Independent	
	c. Exclusive	d. None of these.	
14	The variance of exponential(λ) distribution is, select the correct option		3
	a. $1/\lambda$	b. $1/\lambda^2$	
	c. $1/\lambda(1-\lambda)$	d. $(1-\lambda)/\lambda$	
15.	Which of the following discrete distribution follows “lack of memory property”, Select the correct option		3
	a. Bernoulli distribution	b. Binomial distribution	
	c. Geometric distribution	d. Poisson distribution.	
16.	Select the correct option for the point of inflection for a Normal distribution:		
	a. $x+\sigma$	b. $x-\sigma$	

	c. $X \pm \sigma$	d. $X+2\sigma$
17.	Identify the mode of a Poisson distribution with mean 2.5	
	a. 4	b. 6
	c. 2	d. 7
18.	The skewness in a binomial distribution has a positive value indicates, Select the correct option.	
	a. $p<1/2$	b. $p>1/2$
	c. $p=1/2$	d. $p<q$
19.	Identify the coefficient of variation of Poisson distribution with mean 4 is:	
	a.25%	b. 100%
	c. 50%	d. 75%
20.	The random variables X and Y have variances 0.2 and 0.5 respectively. Let Z= 5X-2Y. Estimate the variance of Z.	
	a.3	b.5
	c.4	d.7
21.	Identify Which of the following statements about the normal distribution is true?	
	a. It is always positively skewed.	b. It is always negatively skewed.
	c. It is symmetric and bell-shaped.	d. None of these
22.	Variance of Binomial distribution (with sample size= n and probability of success= p) is _____. Select the correct option.	
	a. np	b. np(1-p)
	c. n(1-p)	d. None of these
23.	Identify if two random variables X and Y, the relation E(XY)=E(X)E(Y) holds then	
	a. X and Y are statistically independent.	b.For all X and Y
	c. X and Y are identical	d.None of these
24.	If Var(X)=1, then Var($2X \pm 3$) is,select the correct option	
	a. 5	b 13
	c. 4	d.10
25.	The value of $F(\infty)$ is,select the correct option.	

	a. 0	b. 1
	c. -1	d. ± 1
26.	A R.V X takes values 1 and 2 with probability 2/3 and 1/3 select the value of E(X)	
	a.1	b.4/3
	c.2/3	d.0
27.	Let X be a random variable, Then $F(X)=k(1-e^{-x})^2$, $x \geq 0$ 0, otherwise Is a distribution function, Identify the value of k.	
	a.-1	b.1/2
	c.1	d.2
28.	Let X and Y be two random variables with E(X)=1, Var(X)=4, and E(Y)=2, Var(Y)=1. Identify the maximum possible value for E[XY]	
	a.3	b.4
	c.2	d.5
29.	If two random variables are independent, their joint probability distribution function is, Select the correct option.	
	a.The sum of their individual probability distribution functions.	b.The product of their individual probability distribution functions.
	c.The difference of their individual probability distribution functions.	d.The ratio of their individual probability distribution functions.
30.	If event A has a probability of 0.3 and event B has a probability of 0.6, estimate the probability of both events A and B occurring if they are independent?	
	a. 0.18	b. 0.3
	c. 0.9	d. 0.36
31	A coin is tossed 5 times. Calculate the probability that all five tosses show heads.	
	a. 0.03	b. 0.01

	c. 0.07	d. None of these
32	A coin is tossed 5 times. Calculate the probability that all five tosses show heads given that at least two coin shows head.	
	a. 0.01	b. 0.03
	c. 0.9999	d.None of these
33	Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, Calculate the probability that the notebook was supplied by A.	
	a. 44/69	b. 25/69
	c.31/69	d. 67/69
34	The mean deviation of Normal distribution is. Choose the correct option.	
	a. $\sigma\sqrt{\frac{2}{\pi}}$	b. $\sigma\sqrt{\frac{\pi}{2}}$
	c. $\sqrt{2\pi}$	d. $\sigma\sqrt{\pi}$
35	For a normal distribution, Q.D: S.D is approximately, choose the correct option	
	a. 67:100	b.80:100
	c. 63:100	d. None of these.
36	If X and Y are random variables with expectations 3 and 5 respectively, then Calculate the expectation of (3X-5Y+16)	
	a.16	b.-16
	c. -2	d. 0
37	The variance of an exponential distribution with mean 1/8 is computed as , select the correct option.	
	a. 1/32	b. 1/16
	c. 1/64	d. 1/24

38.	If X follows a Bin (6,0.45) distribution, Y follows N (0,6) distribution and moreover they are independent, then E(XY) is calculated as, select the correct option.	
	a. 2	b. 5
	c. 4	d. 0
39.	The cumulative distribution function of X following a Bin (4,0.5) distribution at x = 2 is calculated as	
	a. 0.6485	b. 0.6875
	c. 0.6735	d. 0.5885
40	The cumulative distribution function of X following a U (2,4) distribution at x = 2.5 is estimated as	
	a. 0.5	b. 0.25
	c. 0.75	d. 0.65
41	The probability density function of X following an exponential (2) distribution at x = 1 is estimated as	
	a. 0.2515	b. 0.2125
	c. 0.2707	d. 0.1665
42	Which of the following is a discrete random variable, select the correct option	
	a.Height of students in a class	b. Time taken to complete an exam
	c. Number of defective bulbs in a box	d. Weight of a newborn baby
43	A random variable X has probability distribution: X 0 1 2 P(X) 0.2 0.5 0.3 The expected value E(X) is, select the correct option	
	a.1.0	b.1.1
	c.1.2	d.1.3
44	If E(X)=5 and $E(X^2)$=31, then Var(X) is, select the correct option	

	a.5	b.6
	c.25	d.31
45	If X~Bin(10,0.4), then the mean of X is, select the correct option	
	a.2	b.4
	c.6	d.10

Short Answer Type Question:

1	Discuss probability mass function(PMF) and it's properties.
	Ans: For a discrete random variable X, there exist a function f(x) such that, f(x)=P(X=x), x being the general value of X. A function of this type satisfying the conditions I) f(x)≥ 0 for all x II) $\sum f(x) = 1$
2	Discuss probability density function(PDF) and it's properties.
	Ans: : For a Continuous random variable X, there exist a function f(x) such that, $\int_a^b f(x) = P(a \leq X \leq b)$, x being the general value of X. A function of this type satisfying the conditions I) f(x)≥ 0 for all x II) $\int_{-\infty}^{\infty} f(x)dx = 1$
3	If a random variable X has the pdf f(x)=1/4,-2<x<2 0, elsewhere Identify the value of P{(2x+3)>5}.
	Ans: we have find the probability of P{(2x+3)>5} =P(2x>5-3) =P(2x>2) =P(x>1) Now P(x>1) = $\int_1^2 f(x)dx$ $= \int_1^2 \frac{1}{4}\, dx = \frac{1}{4}\int_1^2 dx = \frac{1}{4}[x]_1^2 = \frac{1}{4}$
4	Show that mean of Bin(n,p) distribution is np.

	$$E(X) \overset{(1)}{=} \sum_{x=0}^{n} x \times \binom{n}{x} p^x (1-p)^{n-x}$$ $$\overset{(2)}{=} \sum_{x=1}^{n} x \times \binom{n}{x} p^x (1-p)^{n-x}$$ $$\overset{(3)}{=} \sum_{x=1}^{n} x \times \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$ $$= \sum_{x=1}^{n} \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x}$$ $$\overset{(4)}{=} np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{n-x}$$ $$\overset{(5)}{=} np \sum_{y=0}^{n-1} \frac{(n-1)!}{y!(n-y-1)!} p^y (1-p)^{n-y-1}$$ $$\overset{(6)}{=} np \left(\sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{n-y-1} \right)$$ $$\overset{(7)}{=} np \times 1$$ $$= np;$$
5	Estimate the mean of a Uniform (a, b) distribution.
	$$E(X) - \int_a^b x \cdot \frac{1}{b-a} \, dx$$ $$= \left[\frac{1}{2} \frac{x^2}{b-a} \right]_a^b$$ $$= \frac{1}{2} \frac{b^2 - a^2}{b-a}$$ $$= \frac{1}{2} \frac{(b+a)(b-a)}{b-a}$$ $$= \frac{1}{2}(a+b) .$$
6	Estimate the value of mean for poisson (λ) distribution.

$$\begin{aligned}\mathrm{E}(X) &= \sum_{x=0}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \sum_{x=1}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} \cdot \sum_{x=1}^{\infty} \frac{x}{x!} \lambda^x \\ &= \lambda e^{-\lambda} \cdot \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \,.\end{aligned}$$

Substituting $z = x - 1$, such that $x = z + 1$, we get:

$$\mathrm{E}(X) = \lambda e^{-\lambda} \cdot \sum_{z=0}^{\infty} \frac{\lambda^z}{z!} \,.$$

Using the power series expansion of the exponential function

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \,,$$

the expected value of X finally becomes

$$\begin{aligned}\mathrm{E}(X) &= \lambda e^{-\lambda} \cdot e^{\lambda} \\ &= \lambda \,.\end{aligned}$$

7	A random variable X follows binomial distribution with mean 5/3 and P(X=2)=P(X=1). Estimate the mode and variance of the distribution.
	Hint: E(X)= np=$\frac{5}{3}$ P(X=2)=P(x=2) ${}^{n}c_2\, p^2\, (1\text{-}p)^{n-2} = {}^{n}c_1\, p^1\, (1\text{-}p)^{n-1}$ $p = \frac{1}{3}$, $q = \frac{2}{3}$ n=5 mode= (n+1)p=2 Hence the modes are, (1,2) Variance is V(X)= npq = $5 * \frac{1}{3} * \frac{2}{3} = \frac{10}{9}$
8	A random variable X has the following probability distribution: X | 0 | 1 | 2 | 3 $P(X = x)$ | k | $2k$ | $3k$ | $4k$ Calculate (i) the value of k, (ii) E(X).

Since probabilities sum to 1,

$$k + 2k + 3k + 4k = 1$$

$$10k = 1$$

$$\boxed{k = \frac{1}{10}}$$

Now,

$$E(X) = \sum xP(x)$$

$$= 0\left(\frac{1}{10}\right) + 1\left(\frac{2}{10}\right) + 2\left(\frac{3}{10}\right) + 3\left(\frac{4}{10}\right)$$

$$= \frac{2 + 6 + 12}{10} = \frac{20}{10} = 2$$

Answer:

$$\boxed{k = \frac{1}{10}, \qquad E(X) = 2}$$

9	Describe Cumulative Distribution function and its properties.

A Cumulative Distribution Function (CDF) is a mathematical concept used in probability theory and statistics to describe the probability distribution of a random variable. The CDF provides information about the probability that a random variable takes on a value less than or equal to a given point.

For a random variable X, the CDF, denoted as F(x), is defined as:

$$F(x) = P(X \leq x)$$

Here are some properties of the Cumulative Distribution Function:

1. **Non-decreasing**: The CDF is a non-decreasing function. This means that as the value of x increases, the CDF either remains constant or increases but never decreases.
2. **Right-Continuous**: The CDF is right-continuous, meaning that the function has no jumps or discontinuities from the right side. Mathematically, this is expressed as $\lim_{h \to 0^+} F(x+h) = F(x)$
3. **Bounded**: The CDF is bounded between 0 and 1. This is because probabilities are always between 0 and 1, and the CDF represents cumulative probabilities.
4. **Limits at Infinity**: As x approaches negative infinity, $F(x)$ approaches 0, and as x approaches positive infinity, $F(x)$ approaches 1.

$$\lim_{x \to -\infty} F(x) = 0$$

$$\lim_{x \to \infty} F(x) = 1$$

10 A perfect coin is tossed 3 time in succession. Given X=1 if first toss gives head, X=0 if first toss gives tail, and Y=number of heads obtained in 3 tosses, Describe the joint distribution of x and y.

Hint: X=1 if first toss gives head
X=0 if 1st toss gives tail.
Y=number of head obtained in 3 tosses
Joint distribution table,

X \ Y	0	1	2	3
0	1/8	2/8	1/8	0
1	0	1/8	2/8	1/8

11 A machine produces items, each of which is defective with probability 0.1.If 8 items are selected at random, estimate the probability that at most two items are defective.

Let X denote the number of defective items among the 8 selected.

Since each item is defective with probability $p = 0.1$ and selections are independent,
$X \sim \text{Binomial}(n = 8, p = 0.1)$.

The required probability is

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2).$$

$$P(X = 0) = \binom{8}{0}(0.1)^0(0.9)^8 = (0.9)^8 = 0.4305$$

$$P(X = 1) = \binom{8}{1}(0.1)(0.9)^7 = 8(0.1)(0.9)^7 = 0.3826$$

$$P(X = 2) = \binom{8}{2}(0.1)^2(0.9)^6 = 28(0.01)(0.9)^6 = 0.1488$$

Adding,

$$P(X \leq 2) = 0.4305 + 0.3826 + 0.1488 = 0.9619 \text{ (approx)}$$

$$\boxed{P(\text{at most 2 defective items}) \approx 0.962}$$

12	The number of typing errors per page in a document follows a Poisson distribution with mean 1.5. Estimate the probability that a randomly selected page has(i)No errors, and (ii)At least two errors.

Let X denote the number of typing errors on a randomly selected page.

Given that the number of errors follows a Poisson distribution with mean $\lambda = 1.5$, we have

$$P(X = x) = \frac{e^{-1.5}(1.5)^x}{x!}, \quad x = 0, 1, 2, \ldots$$

(i) Probability of no errors

$$P(X = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5}$$

$$P(X = 0) \approx 0.2231$$

(ii) Probability of at least two errors

$$P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$$

$$P(X = 1) = \frac{e^{-1.5}(1.5)^1}{1!} = 1.5e^{-1.5}$$

$$P(X \geq 2) = 1 - \left[e^{-1.5} + 1.5e^{-1.5}\right]$$

$$P(X \geq 2) = 1 - 2.5e^{-1.5}$$

$$P(X \geq 2) \approx 1 - 0.5578 = 0.4422$$

13	A machine produces items, each of which is defective with probability 0.1. If 8 items are selected at random, estimate the probability that at most two items are defective.

Each item produced by the machine is defective with probability

$$p = 0.1$$

Hence, the probability that an item is non-defective is

$$q = 1 - p = 0.9$$

Let X denote the number of defective items in a random sample of 8 items.
Then X follows a binomial distribution with parameters $n = 8$ and $p = 0.1$.

$$P(X = x) = \binom{8}{x}(0.1)^x(0.9)^{8-x}$$

We are required to find

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$P(X = 0) = \binom{8}{0}(0.1)^0(0.9)^8 = (0.9)^8$$

$$P(X = 1) = \binom{8}{1}(0.1)(0.9)^7$$

$$P(X = 2) = \binom{8}{2}(0.1)^2(0.9)^6$$

Therefore,

$$P(X \leq 2) = (0.9)^8 + 8(0.1)(0.9)^7 + 28(0.01)(0.9)^6$$

$$= 0.4305 + 0.3826 + 0.1488$$

$$= 0.9619 \text{ (approx)}$$

14	The number of phone calls received by a call center follows a Poisson distribution with an average of 4 calls per hour. Estimate is the probability that exactly 2 calls are received in an hour.

We are given:

- The number of calls follows a **Poisson distribution**
- Mean (λ) = 4 calls per hour
- We want: $P(X = 2)$

Poisson Probability Formula:

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

Substitute $\lambda = 4, k = 2$:

$$P(X = 2) = \frac{e^{-4} \cdot 4^2}{2!} = \frac{e^{-4} \cdot 16}{2} = 8 \cdot e^{-4}$$

Now compute:

$$e^{-4} \approx 0.0183$$

$$P(X = 2) = 8 \cdot 0.0183 = 0.1464$$

15 A fair coin is tossed 10 times. Calculate the probability of obtaining at least 7 heads

A fair coin is tossed 10 times.

Find

$$P(X \geq 7)$$

Here,

$$n = 10, \quad p = \frac{1}{2}$$

$$P(X \geq 7) = P(7) + P(8) + P(9) + P(10)$$

$$= \frac{\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{2^{10}}$$

$$= \frac{120 + 45 + 10 + 1}{1024}$$

$$= \frac{176}{1024}$$

$$= \boxed{0.1719}$$

16 The heights of students are normally distributed with mean 170 cm and standard deviation 8 cm. Calculate the probability that a randomly selected student has height greater than 182 cm.(Given: P(Z>1.5)=0.0668)

Normal distribution

$$\mu = 170, \qquad \sigma = 8$$

Find

$$P(X > 182)$$

Standardize,

$$Z = \frac{182 - 170}{8} = \frac{12}{8} = 1.5$$

Given,

$$P(Z > 1.5) = 0.0668$$

Hence,

$$\boxed{P(X > 182) = 0.0668}$$

17 The random variable X has the probability density function

$$f(x) = \begin{cases} \frac{1}{6}, & 2 < x < 8, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate:-(i) P(3<X<6). (ii) The mean of the distribution.

Given

$$f(x) = \begin{cases} \frac{1}{6}, & 2 < x < 8 \\ 0, & \text{otherwise} \end{cases}$$

Clearly,

$$X \sim U(2, 8)$$

(i)

$$P(3 < X < 6) = \frac{6 - 3}{8 - 2} = \frac{3}{6} = \boxed{\frac{1}{2}}$$

(ii)

Mean

$$E(X) = \frac{2 + 8}{2} = 5$$

Answer

$$\boxed{P(3 < X < 6) = \frac{1}{2}, \qquad E(X) = 5}$$

18 Customers arrive at a service counter according to a Poisson distribution with mean 4 customers per hour. Complete the probability that exactly five customers arrive in one hour.

Mean

$$\lambda = 4$$

Find

$$P(X = 5)$$

$$= \frac{e^{-4}4^5}{5!}$$

Now,

$$4^5 = 1024, \qquad 5! = 120$$

$$e^{-4} = 0.0183$$

Hence,

$$P(X = 5) = \frac{0.0183 \times 1024}{120}$$

$$= \boxed{0.156}$$

Long Answer Type Question:

1	The probability density function of a continuous distribution is given by $f(x)=\frac{3}{4}x(2-x), 0<x<2.$ Evaluate the mean and P(X<1)

	Great! Now that we have the probability density function (PDF) $f(x) = \frac{3}{4}(2 - x)$ defined on the interval $0 < x < 2$, we can proceed to calculate the mean (μ) and $P(X < 1)$. **Mean (μ):** The mean of a continuous distribution is given by the integral: $\mu = \int x \cdot f(x)\, dx$ For this PDF: $\mu = \int_0^2 x \cdot \frac{3}{4}(2 - x)\, dx$ You can calculate this integral to find the mean. **Probability $P(X < 1)$:** The probability $P(X < 1)$ is given by the integral: $P(X < 1) = \int_0^1 f(x)\, dx$ For this PDF: $P(X < 1) = \int_0^1 \frac{3}{4}(2 - x)\, dx$ Calculate this integral to find $P(X < 1)$. If you perform the integrations, you will obtain the mean (μ) and the probability $P(X < 1)$.
2	Show that the variance of Binomial(n,p) distribution is npq.

The Binomial distribution with parameters n and p describes the number of successes in n independent Bernoulli trials, where each trial has a probability p of success and $q = 1 - p$ of failure.

The variance of a random variable X can be calculated using the formula:

$$Var(X) = E(X^2) - (E(X))^2$$

For a Binomial distribution, the mean $E(X)$ is np, and the expectation $E(X^2)$ can be calculated using the formula for the second moment:

$$E(X^2) = \sum_{k=0}^{n} k^2 \cdot P(X = k)$$

where $P(X = k)$ is the probability mass function of the Binomial distribution, given by:

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot q^{n-k}$$

Now, let's calculate the variance:

$$Var(X) = E(X^2) - (E(X))^2$$

$$Var(X) = \sum_{k=0}^{n} k^2 \cdot P(X = k) - (np)^2$$

$$Var(X) = \sum_{k=0}^{n} k^2 \cdot \binom{n}{k} \cdot p^k \cdot q^{n-k} - (np)^2$$

After calculating this expression, you'll find that it simplifies to npq, which demonstrates that the variance of a Binomial distribution with parameters n and p is npq.

3	Show that the mean of Normal distribution(μ,σ^2) is μ.

	The mean of a continuous random variable X with probability density function (PDF) $f(x)$ is given by the expected value: $E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$ For a normal distribution with mean μ and variance σ^2, the probability density function is given by: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ Now, let's calculate the expected value: $E(X) = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx$ This integral involves the Gaussian function, and when you perform the integration, you will find that the result is μ, the mean of the normal distribution. Therefore, the mean of a normal distribution with parameters μ and σ^2 is μ.
4	Illustrate the mean and variance of Uniform distribution.

	The Uniform distribution, denoted as $U(a, b)$, is a continuous probability distribution with a constant probability density function (PDF) between the parameters a and b and zero elsewhere. The probability density function (PDF) of a Uniform distribution $U(a, b)$ is given by: $f(x) = \frac{1}{b-a}$ where a and b are the parameters defining the range of the distribution. **Mean of Uniform Distribution:** The mean (μ) of a continuous uniform distribution is calculated as the average of the minimum (a) and maximum (b) values: $\mu = \frac{a+b}{2}$ **Variance of Uniform Distribution:** The variance (σ^2) of a continuous uniform distribution is calculated using the formula: $\sigma^2 = \frac{(b-a)^2}{12}$ These formulas demonstrate that the mean is the midpoint of the interval, and the variance is related to the length of the interval ($b - a$). In summary, for a Uniform distribution $U(a, b)$: • Mean (μ): $\frac{a+b}{2}$ • Variance (σ^2): $\frac{(b-a)^2}{12}$
5	Show that the mean deviation about mean for Normal distribution is σ √2/∏.

density function (PDF) is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Now, let's calculate the mean deviation about the mean:

$$E\left(|X-\mu|\right) = \int_{-\infty}^{\infty} |x-\mu| \cdot f(x)\, dx$$

Substitute the PDF of the normal distribution into the integral:

$$E\left(|X-\mu|\right) = \int_{-\infty}^{\infty} |x-\mu| \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\, dx$$

To simplify this integral, we can use the symmetry of the normal distribution. The integral from $-\infty$ to μ is equal to the integral from μ to ∞, so we can consider only the positive part:

$$2\int_{\mu}^{\infty} (x-\mu) \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\, dx$$

Now, make the substitution $u = \frac{x-\mu}{\sigma}$ to simplify the integral:

$$2\int_{0}^{\infty} u \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)\, du$$

The integral involving u is an integral of the standard normal distribution, and its expected value is $\sqrt{\frac{2}{\pi}}$. Therefore:

$$E\left(|X-\mu|\right) = 2 \cdot \sqrt{\frac{2}{\pi}} \cdot \sigma$$

Finally, substitute this result into the formula for mean deviation about the mean:

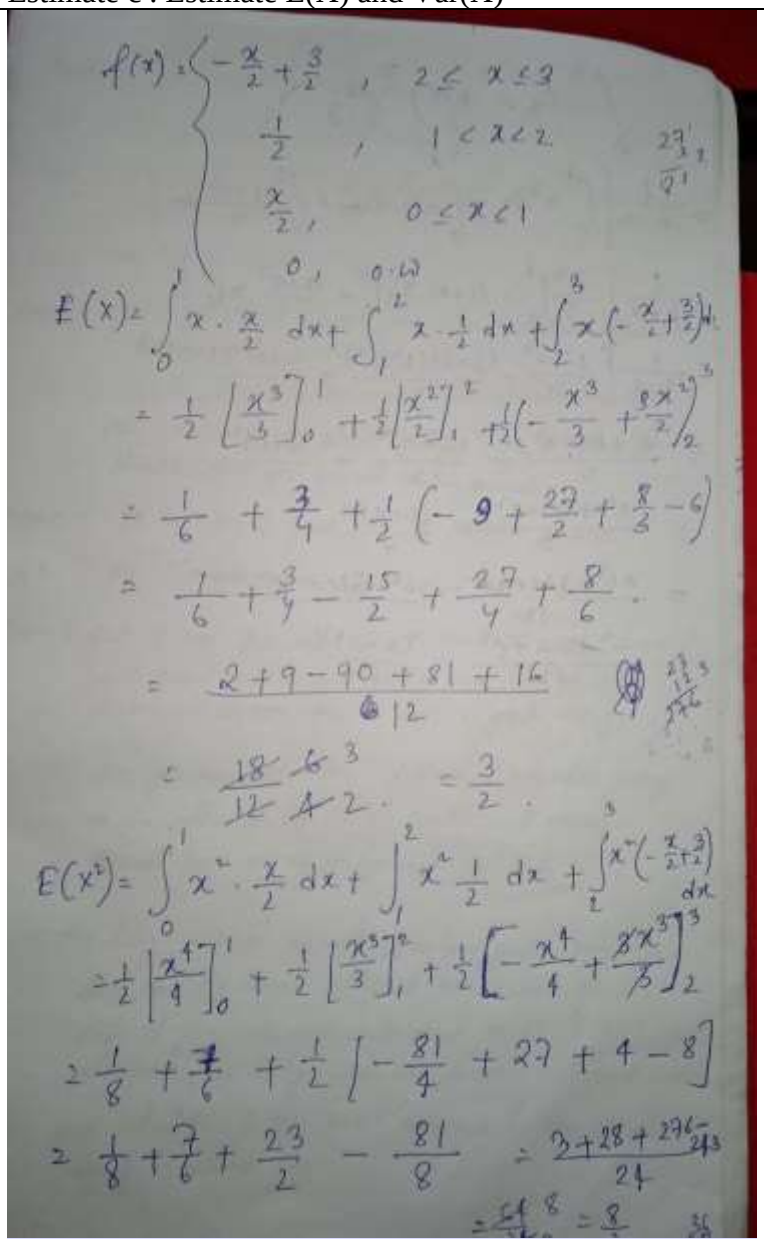
$$\text{Mean Deviation about Mean} = 2 \cdot \sqrt{\frac{2}{\pi}} \cdot \sigma$$

This demonstrates that the mean deviation about the mean for a normal distribution is $\sigma \cdot \sqrt{\frac{2}{\pi}}$.

6	The probability density function of a continuous random variable is given by

$$f(x)=\begin{cases}-cx+3c, & 2\le x\le 3\\ c, & 1<x<2\\ cx, & 0\le x\le 1\\ 0, & \text{elsewhere}\end{cases}$$

Estimate c . Estimate E(X) and Var(X)



7	Show that the mean and variance of Poisson distribution are equal.

For a Poisson distribution with parameter λ, the probability mass function (PMF) is given by:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

where k is a non-negative integer representing the number of events.

The mean ($E(X)$) of a Poisson distribution is calculated as:

$$E(X) = \sum_{k=0}^{\infty} k \cdot P(X = k)$$

$$E(X) = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!}$$

Simplify this expression:

$$E(X) = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}$$

Now, substitute $j = k - 1$:

$$E(X) = e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+1}}{j!}$$

$$E(X) = e^{-\lambda}\lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

The sum in the expression is the Taylor series expansion of e^{λ}:

$$E(X) = e^{-\lambda}\lambda \cdot e^{\lambda}$$

$$E(X) = \lambda$$

Now, let's calculate the variance ($Var(X)$) of the Poisson distribution:

$$Var(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot P(X = k)$$

$$E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot \frac{e^{-\lambda}\lambda^k}{k!}$$

Simplify this expression:

$$E(X^2) = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}$$

Now, substitute $j = k - 2$:

$$E(X^2) = e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+2}}{j!}$$

$$E(X^2) = e^{-\lambda}\lambda^2 \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

The sum in the expression is the Taylor series expansion of e^{λ}:

$$E(X^2) = e^{-\lambda}\lambda^2 \cdot e^{\lambda}$$

$$E(X^2) = \lambda^2 + \lambda$$

Now, calculate $Var(X)$:

$$Var(X) = E(X^2) - (E(X))^2$$

$$Var(X) = (\lambda^2 + \lambda) - \lambda^2$$

$$Var(X) = \lambda$$

8	Show that Poisson distribution may be regarded as a limiting form of the binomial distribution when number of trials n->∞ and p->0

The Poisson Distribution may be regarded as a limiting form of the binomial distribution when number of trials $n \to \infty$, and probability of success in trial $p \to 0$ but $np(=\mu)$ remains finite.

$$\lim_{n\to\infty} \binom{n}{r} \left(\frac{\mu}{n}\right)^r \left(1-\frac{\mu}{n}\right)^{n-r}$$

$$= \lim_{n\to\infty} \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} \left(\frac{\mu}{n}\right)^r \left(1-\frac{\mu}{n}\right)^{n-r}$$

$$= \lim_{n\to\infty} \frac{1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right)}{r!} (\mu)^r \left(1-\frac{\mu}{n}\right)^{n-r}$$

$$= \frac{\mu^r}{r!} \cdot \lim_{n\to\infty} \left[1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right)\right] \cdot \lim_{n\to\infty} \left(1-\frac{\mu}{n}\right)^n \cdot \frac{1}{\lim_{n\to\infty}\left(1-\frac{\mu}{n}\right)^r}$$

$$= \lim_{n\to\infty} \left(1-\frac{\mu}{n}\right)^n \cdot \frac{\mu^r}{r!} \cdot \lim_{n\to\infty} \left[1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right)\right] \cdot \frac{1}{\lim_{n\to\infty}\left(1-\frac{\mu}{n}\right)^r}$$

$$= e^{-\mu} \cdot \frac{\mu^r}{r!} \cdot 1 \cdot \frac{1}{1}$$

$$= \frac{e^{-\mu}\mu^r}{r!}.$$

The practical utility of the result is that, when n is sufficiently large and p is quite small, but $np(=\mu)$ has a moderate value, the binomial probabilities $\binom{n}{r}\left(\frac{\mu}{n}\right)^r\left(1-\frac{\mu}{n}\right)^{n-r}$ can be well approximated by the corresponding Poisson Probabilities $\frac{e^{-\mu}\mu^r}{r!}$, which is much easier to compute.

9	The mean yield for one acre plot is 662kilos with a s.d 32kilos.Assuming normal distribution, estimate how many one acre plots in a batch of 1000 plots would you expect to have yield i) over 700 kilos, ii)below 650 kilos.

i) For yields over 700 kilos:

$P(X > 700) = 1 - P(X < 700)$

ii) For yields below 650 kilos:

$P(X < 650)$

Once we have the proportions, we can multiply them by 1000 to find the estimated number of plots.

Let's calculate:

i) For yields over 700 kilos:

$P(X > 700) = 1 - P(X < 700) = 1 - P(Z < 1.1875)$

Looking up 1.1875 in the standard normal distribution table, we find that $P(Z < 1.1875) \approx 0.882$.

So, $P(X > 700) \approx 1 - 0.882 = 0.118$.

Estimated number of plots over 700 kilos = $0.118 \times 1000 = 118$ plots.

ii) For yields below 650 kilos:

$P(X < 650) = P(Z < -0.375)$

Looking up -0.375 in the standard normal distribution table, we find that $P(Z < -0.375) \approx 0.352$.

So, $P(X < 650) \approx 0.352$.

Estimated number of plots below 650 kilos = $0.352 \times 1000 = 352$ plots.

10 The following table gives the joint distribution of x and y:

X / Y	0	1	2
1	0.3	0.2	0.1
2	0.1	0.0	0.3

Are x and y independent, also identify the correlation coefficient between x and y.

To check whether X and Y are independent, we need to verify if the joint probability distribution can be expressed as the product of the marginal probability distributions.

The marginal probability distributions for X and Y can be calculated by summing the probabilities across the corresponding rows and columns:

X/Y	0	1	2	Marginal (P(X=x))
1	0.3	0.2	0.1	0.6
2	0.1	0.0	0.3	0.4
Marginal (P(Y=y))	0.4	0.2	0.4	1.0

If X and Y are independent, then the joint distribution $P(X = x, Y = y)$ should equal the product of the marginal distributions $P(X = x) \cdot P(Y = y)$ for all possible values of x and y.

Let's compare:

$$P(X = 1, Y = 0) = 0.3 \neq (0.6 \cdot 0.4)$$
$$P(X = 1, Y = 1) = 0.2 \neq (0.6 \cdot 0.2)$$
$$P(X = 1, Y = 2) = 0.1 \neq (0.6 \cdot 0.4)$$
$$P(X = 2, Y = 0) = 0.1 \neq (0.4 \cdot 0.4)$$
$$P(X = 2, Y = 1) = 0.0 \neq (0.4 \cdot 0.2)$$
$$P(X = 2, Y = 2) = 0.3 \neq (0.4 \cdot 0.4)$$

Since, for some values of X and Y, $P(X = x, Y = y)$ does not equal $P(X = x) \cdot P(Y = y)$, X and Y are not independent.

Now, let's find the correlation coefficient between X and Y. The formula for the correlation coefficient (ρ) is given by:

$$\rho = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

For discrete random variables, the covariance ($\text{Cov}(X,Y)$) can be calculated as:

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$\text{Cov}(X,Y) = \sum_x \sum_y xy \cdot P(X = x, Y = y) - E(X)E(Y)$$

First, calculate the expected values:

$$E(X) = 1 \cdot 0.6 + 2 \cdot 0.4 = 0.6 + 0.8 = 1.4$$

$$E(Y) = 0 \cdot 0.4 + 1 \cdot 0.2 + 2 \cdot 0.4 = 0 + 0.2 + 0.8 = 1$$

$$E(XY) = 1 \cdot 0 \cdot 0.3 + 1 \cdot 1 \cdot 0.2 + 1 \cdot 2 \cdot 0.1 + 2 \cdot 0 \cdot 0.1 + 2 \cdot 1 \cdot 0.0 + 2 \cdot 2 \cdot 0.3 = 0.2 + 0.2 + 0.2 + 0 + 0 + 1.2 = 1.8$$

Now, substitute these values into the covariance formula:

$$\text{Cov}(X,Y) = 1.8 - (1.4 \cdot 1) = 1.8 - 1.4 = 0.4$$

Next, calculate the variances:

Next, calculate the variances:

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2$$

$$E(X^2) = 1^2 \cdot 0.6 + 2^2 \cdot 0.4 = 0.6 + 1.6 = 2.2$$

$$E(Y^2) = 0^2 \cdot 0.4 + 1^2 \cdot 0.2 + 2^2 \cdot 0.4 = 0 + 0.2 + 1.6 = 1.8$$

$$\text{Var}(X) = 2.2 - (1.4)^2 = 2.2 - 1.96 = 0.24$$

$$\text{Var}(Y) = 1.8 - (1)^2 = 1.8 - 1 = 0.8$$

11 Define a probability distribution of a discrete random variable. A random variable X has the following probability distribution:

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

(i) Verify whether this is a valid probability distribution.
(ii) Calculate the mean and variance of X.

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Sum of probabilities:

$$0.1 + 0.3 + 0.4 + 0.2 = 1.0$$

All probabilities are between 0 and 1.

Let $\mu = E(X) = \sum X \cdot P(X)$

$$E(X) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 0 + 0.3 + 0.8 + 0.6 = 1.7$$

Now calculate $E(X^2) = \sum X^2 \cdot P(X)$

$$E(X^2) = 0^2(0.1) + 1^2(0.3) + 2^2(0.4) + 3^2(0.2) = 0 + 0.3 + 1.6 + 1.8 = 3.7$$

Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 3.7 - (1.7)^2 = 3.7 - 2.89 = 0.81$$

12 Let X be a continuous random variable with probability density function (pdf):
f(x) = 2x for $0 \leq x \leq 1$,
and f(x) = 0 elsewhere.

	(i) Test whether this is a valid pdf. (ii) Also find E(X) and Var(X).
	A function $f(x)$ is a valid pdf if: 1. $f(x) \geq 0$ for all x 2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$ Check the integral: $$\int_0^1 2x\,dx = [x^2]_0^1 = 1^2 - 0 = 1$$ $$E(X) = \int_0^1 x \cdot f(x)\,dx = \int_0^1 x \cdot 2x\,dx = \int_0^1 2x^2\,dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$$ **Find $E(X^2)$:** $$E(X^2) = \int_0^1 x^2 \cdot f(x)\,dx = \int_0^1 x^2 \cdot 2x\,dx = \int_0^1 2x^3\,dx = \left[\frac{2x^4}{4}\right]_0^1 = \frac{2}{4} = \frac{1}{2}$$ **Variance:** $$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9-8}{18} = \frac{1}{18}$$
13	The probability that a person receives at least one phone call in an hour is 0.8.Assuming that the number of calls follows a Poisson distribution, estimate the mean number of calls per hour.

Let X denote the number of phone calls received in an hour.

It is given that X follows a Poisson distribution with mean λ.

Probability of receiving at least one call in an hour is

$$P(X \geq 1) = 0.8$$

Now,

$$P(X \geq 1) = 1 - P(X = 0)$$

Hence,

$$1 - P(X = 0) = 0.8$$

$$P(X = 0) = 0.2$$

For a Poisson distribution,

$$P(X = 0) = e^{-\lambda}$$

Therefore,

$$e^{-\lambda} = 0.2$$

Taking natural logarithm on both sides,

$$-\lambda = \ln(0.2)$$

$$\lambda = -\ln(0.2)$$

$$\lambda = \ln(5) \approx 1.609$$

14	The weights of packets of rice are normally distributed with mean 1 kg and standard deviation 0.05 kg.Estimate the probability that a randomly selected packet weighs(i)between 0.95 kg and 1.05 kg.(ii) Less than 0.95 kg.

Let X denote the weight (in kg) of a packet of rice.

Given

Mean $\mu = 1$ kg

Standard deviation $\sigma = 0.05$ kg

Then

$$X \sim N(1,\ 0.05^2)$$

Standardize using

$$Z = \frac{X - \mu}{\sigma}$$

(i) Probability that the weight lies between 0.95 kg and 1.05 kg

$$P(0.95 < X < 1.05)$$

$$= P\left(\frac{0.95 - 1}{0.05} < Z < \frac{1.05 - 1}{0.05}\right)$$

$$= P(-1 < Z < 1)$$

From standard normal tables,

$$P(-1 < Z < 1) = 0.6826$$

$$\boxed{P(0.95 < X < 1.05) = 0.6826}$$

(ii) Probability that the weight is less than 0.95 kg

$$P(X < 0.95)$$

$$= P\left(Z < \frac{0.95 - 1}{0.05}\right)$$

$$= P(Z < -1)$$

From standard normal tables,

$$P(Z < -1) = 0.1587$$

$$\boxed{P(X < 0.95) = 0.1587}$$

15 If X represents the outcome in rolling a die, calculate F(X).

Interval of x	$F(x) = P(X \leq x)$
$x < 1$	0
$1 \leq x < 2$	$\frac{1}{6}$
$2 \leq x < 3$	$\frac{2}{6} = \frac{1}{3}$
$3 \leq x < 4$	$\frac{3}{6} = \frac{1}{2}$
$4 \leq x < 5$	$\frac{4}{6} = \frac{2}{3}$
$5 \leq x < 6$	$\frac{5}{6}$
$x \geq 6$	1

16 If a random variable X follows binomial distribution with parameter 10 and p and P(X=5)=2P(X=4),calculate P(X>0 | X≤ 2) and P(X=1).

For a binomial distribution,

$$P(X = r) = \binom{10}{r} p^r q^{10-r},$$

where $q = 1 - p$.

Given

$$P(X = 5) = 2P(X = 4).$$

Substitute the formula:

$$\binom{10}{5} p^5 q^5 = 2\binom{10}{4} p^4 q^6.$$

Divide by $p^4 q^5$:

$$252p = 2(210)q.$$

$$252p = 420(1 - p).$$

$$672p = 420.$$

$$\boxed{p = \frac{5}{8}}$$

Hence

$$q = \frac{3}{8}.$$

Step 2: Find $P(X > 0 \mid X < 2)$

Since $X < 2$,

$$X = 0 \text{ or } 1.$$

Also,

$$X > 0 \Rightarrow X = 1.$$

Therefore,

$$P(X > 0 \mid X < 2) = \frac{P(X = 1)}{P(X = 0) + P(X = 1)}.$$

Now,

$$P(X = 0) = \left(\frac{3}{8}\right)^{10} = \frac{59049}{1073741824}.$$

$$P(X = 1) = 10\left(\frac{5}{8}\right)\left(\frac{3}{8}\right)^{9} = \frac{984150}{1073741824}.$$

Thus,

$$P(X > 0 \mid X < 2) = \frac{984150}{984150 + 59049} = \frac{50}{53} \approx 0.9434.$$

$$P(X=1) = 10\left(\frac{5}{8}\right)\left(\frac{3}{8}\right)^{9}$$

$$\boxed{P(X=1) \approx 0.9166}$$

Final Answer

$$\boxed{P(X>0 \mid X<2) = \frac{50}{53} \approx 0.9434}$$

$$\boxed{P(X=1) \approx 0.9166}$$

17 If a random variable X follows binomial distribution with mean 4 and $E(X^2) = 19$. Calculate P(X=at most 2) and P(X=3).

Mean

$$np = 4.$$

Also,

$$E(X^2) = Var(X) + [E(X)]^2.$$

Thus,

$$19 = npq + 16.$$

$$npq = 3.$$

Since

$$np = 4,$$

$$4q = 3.$$

$$q = \frac{3}{4}, \qquad p = \frac{1}{4}.$$

Now,

$$np = 4$$

gives

$$n\left(\frac{1}{4}\right) = 4.$$

$$\boxed{n = 16}$$

$$P(X \le 2) = P(0) + P(1) + P(2).$$

Now,

$$P(0) = \left(\frac{3}{4}\right)^{16} = 0.01002.$$

$$P(1) = 16\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^{15} = 0.05345.$$

$$P(2) = \binom{16}{2}\left(\frac{1}{4}\right)^{2}\left(\frac{3}{4}\right)^{14} = 0.13363.$$

Hence,

$$P(X \le 2) = 0.01002 + 0.05345 + 0.13363 = \boxed{0.1971}.$$

$$P(3) = \binom{16}{3}\left(\frac{1}{4}\right)^{3}\left(\frac{3}{4}\right)^{13}.$$

$$= 560\left(\frac{1}{64}\right)\left(\frac{3}{4}\right)^{13}$$

$$\boxed{P(X = 3) \approx 0.2079}$$

Final Answer

$$\boxed{P(X \le 2) \approx 0.1971}$$

$$\boxed{P(X = 3) \approx 0.2079}$$

18 The mean of a normally distributed variable X is 10 and P(X>12) = 0.0228, calculate P(X≤ 11).[Given that P(Z≤0.5)=0.6915and P(Z≤ 2) = 0.9772]

Since

$$P(X > 12) = 0.0228,$$

$$P(Z > 2) = 0.0228.$$

Therefore,

$$\frac{12 - 10}{\sigma} = 2.$$

Hence,

$$\boxed{\sigma = 1.}$$

Standardize:

$$Z = \frac{11 - 10}{1} = 1.$$

Therefore,

$$P(X < 11) = P(Z < 1).$$

Using the standard normal table,

$$P(Z < 1) = 0.8413.$$

Final Answer

$$\boxed{P(X < 11) = 0.8413.}$$